An Unconditional VAE for Chest X-Ray Generation

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Abstract

We implement a convolutional variational autoencoder (VAE) in PyTorch and train one unconditional model on pooled NORMAL and PNEUMONIA chest X-rays, following the instructor's clarification. We use a fixed training/validation split and hold out the original test set. The model reaches a stable validation loss and generates recognizable but smooth radiographs. We report training settings, the loss curve, Inception Score (IS), Fréchet Inception Distance (FID), and sample images.

1 Data and method

The dataset contains 5,856 images from Kaggle and the Mendeley collection by Kermany et al. [1]. We used a public mirror that reports unchanged files and splits. After pooling labels, we split the original 5,216-image training folder into 4,694 training and 522 validation images (seed 42). The original 624-image test folder was held out. We did not use the separate 16-image validation folder.

Images were converted to grayscale, resized to 128×128, and scaled to $[0, 1]$. The encoder has five stride-2 convolution blocks (32, 64, 128, 256, and 512 channels). Linear layers predict a 128-dimensional Gaussian latent distribution. A mirrored transposed-convolution decoder produces pixel logits. We minimize summed Bernoulli reconstruction loss plus KL divergence to a standard normal prior.

Table 1: Hyperparameters and selected run settings.

Setting	Value
Input size / channels	128×128 / 1
Latent dimension	128
Optimizer	Adam, learning rate 3×10^{-4}
Batch size	64
Maximum epochs / patience	150 / 18
LR scheduler	Reduce on plateau (patience 6, factor 0.5, min 10^{-5})
Train / split / sample seed	43 / 42 / 2026
Compute	NVIDIA A100-SXM4-80GB, 22.3 min

The best checkpoint was chosen by validation negative ELBO: epoch 120, with loss 9,426.60 per image. Training stopped at epoch 138 after 18 epochs without improvement. A second seed and a 256-dimensional latent variant were also tried; seed 43 had the lowest validation loss (9,426.60 vs. 9,426.81 and 9,427.19). Test scores were not used for selection.

2 Evaluation and results

We use PyTorch-Ignite Inception-v3 metrics [2] with 624 generated samples (seed 2026) and 624 held-out test images.

Preprint.

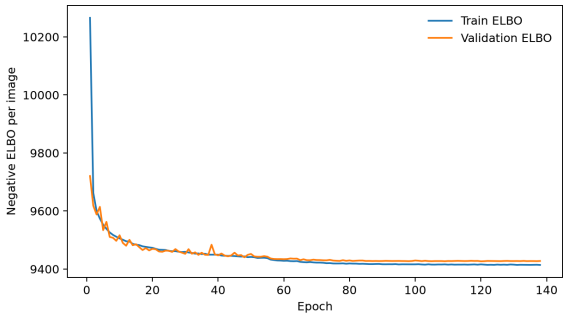
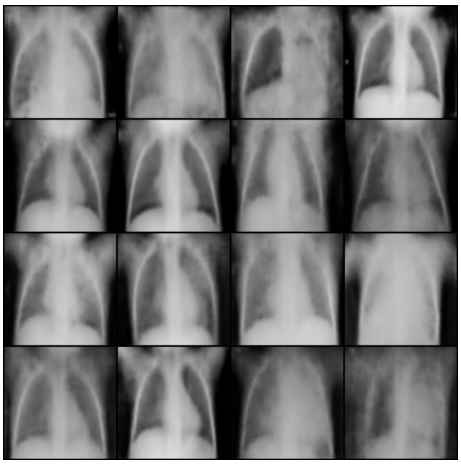


Figure 1: Training and validation negative ELBO per image. The objective falls rapidly at first and then flattens; small later changes are consistent with convergence to a plateau. The epoch-120 checkpoint had the best validation value.



Metric	Result
Inception Score (IS)	1.926
FID	0.162
Images per distribution	624

IS uses generated samples; FID compares test features. Grayscale inputs are repeated as RGB and resized to 299×299. The scores are approximate, not clinical measures.

Figure 2: Sixteen prior samples (left) and held-out test metrics (right). Chest outlines are recognizable, but fine details remain smooth.

3 Analysis

The curve plateaus after rapid early improvement. Samples vary and capture broad chest anatomy, but ribs and lung details are blurred. Square resizing and the 128×128 pixelwise Bernoulli model limit fine detail. Since labels are pooled, the model cannot control whether a sample resembles a NORMAL or PNEUMONIA image.

IS and FID are approximate: Inception is trained on natural images, and FID depends on features and sample count. These 624-sample scores are not clinical measures. The synthetic images must not be used for medical decisions. Code, history, metrics, and all 64 samples accompany this report.

References

[1] D. Kermany, K. Zhang, and M. Goldbaum. Labeled OCT and Chest X-Ray Images for Classification. Mendeley Data, 2018. Dataset: Kaggle; mirror: Hugging Face. CC BY 4.0.

[2] PyTorch-Ignite. GAN evaluation using FID and IS. Tutorial.